A 1 Mohm source: what changes

Bench test 2026-11-06: the same board, firmware and captures as the transient report (Transient_visibility_report.pdf), repeated with a 1 Mohm resistor in series with the Rigol output, standing in for a weak, high-impedance source. The Giga was on USB. Everything below compares the 128 new captures with the 128 from the 50 ohm run.

The short version

- **The source resistance and the input capacitance add a low-pass at 8.67 kHz** (time constant 18.4 us, so the input looks like about 18 pF: board plus wiring). The -3 dB point drops from 10.9 to 6.0 kHz and a step now takes 58 us to rise 10-90 % (was 34 us).
- **Short pulses lose more height.** Below ~15 us the recorded peak is still the pulse's area, but now height x width / 43 us instead of / 16 us: about 2.7x smaller. Nothing looks shorter than ~39 us (was ~12 us).
- **Mains hum appears: 78 mV peak at 60 Hz** (it was under 1 mV with the 50 ohm source). The high-impedance node picks it up. As recorded, it raises the noise at SMOOTH 8 from 11.7 to 57 mV rms, and SMOOTH can't average it away.
- **Remove the hum and the noise is nearly back where it was**: 13.0 mV at SMOOTH 8 and 32.0 mV raw (was 11.7 / 32.6). Little broadband noise is added; the small rise at SMOOTH 8 is low-frequency pickup the 60-300 Hz fit doesn't catch. Hum is slow (16.7 ms per cycle), so over a short transient it is only a drifting baseline.
- **DC is unaffected.** Gain moved -0.036 %, which is inside the run-to-run scatter (up to 0.07 %), so the input resistance is above ~1.5 Gohm. The offset moved -14 mV, which would be about 14 nA of input current through 1 Mohm.

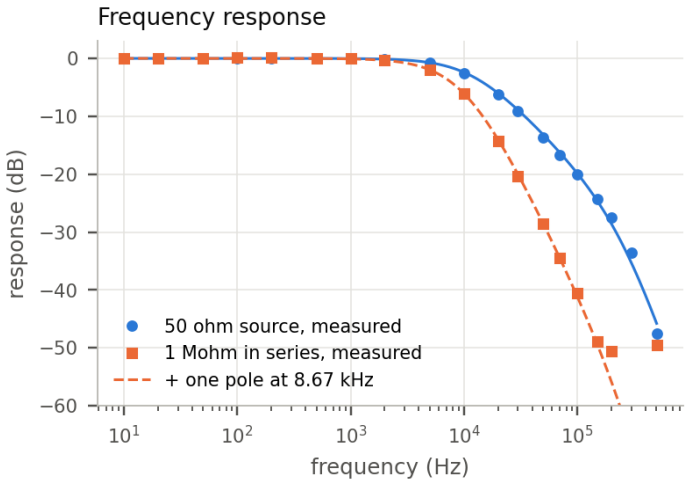
Smallest single-shot pulse at 5 sigma, 500 kS/s SMOOTH 8

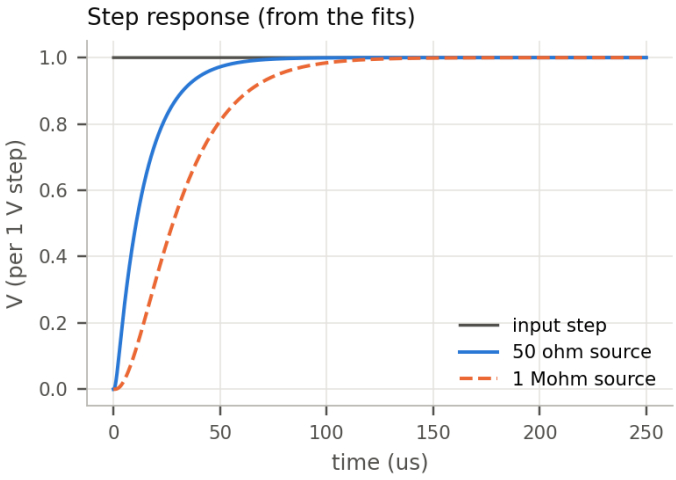
true duration	50 ohm source	1 Mohm, hum removed	1 Mohm, as recorded
1 ms	57 mV	65 mV	283 mV
100 us	57 mV	66 mV	289 mV
10 us	152 mV	297 mV	1301 mV
1 us	1.4 V	2.9 V	13 V

Same method as the main report: 5 x the noise sigma at that setting, divided by the recorded fraction of the height. The response with the extra pole is validated by the captures: the fitted amplitude on the 24 clear 1 Mohm captures has a median of 1.006 (1.00 to 1.02, 5th-95th percentile), against 1.020 for the 50 ohm run. 'As recorded' uses the noise sigma including the hum, so it's the pessimistic case: the hum is a slow baseline, and any local baseline removal recovers most of the 'hum removed' figure.

1. Frequency response and step

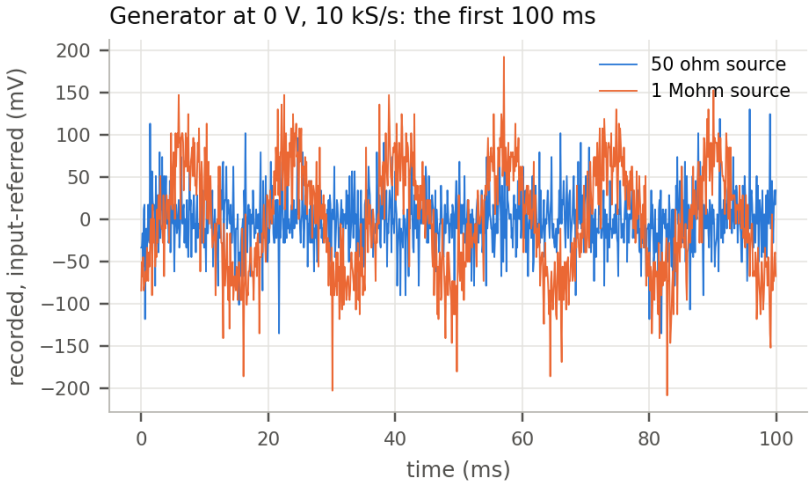
The 1 Mohm sweep (squares) is the 50 ohm sweep times one extra pole at 8.67 kHz; the fit to the difference is within 0.14 dB rms up to 50 kHz. Poles of the 1 Mohm model: 8.7 kHz, 11.7 kHz, 267 kHz, 267 kHz.

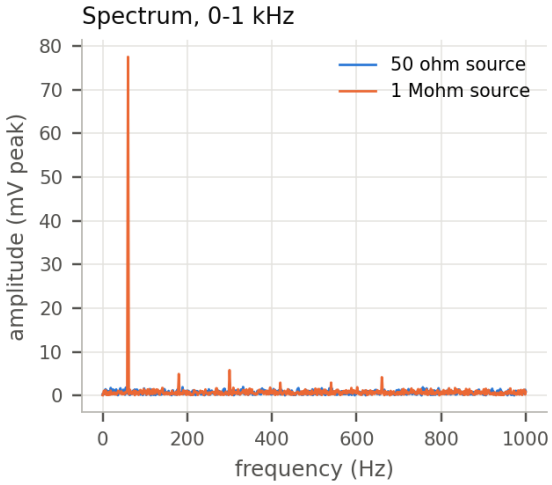




2. Mains hum on a high-impedance input

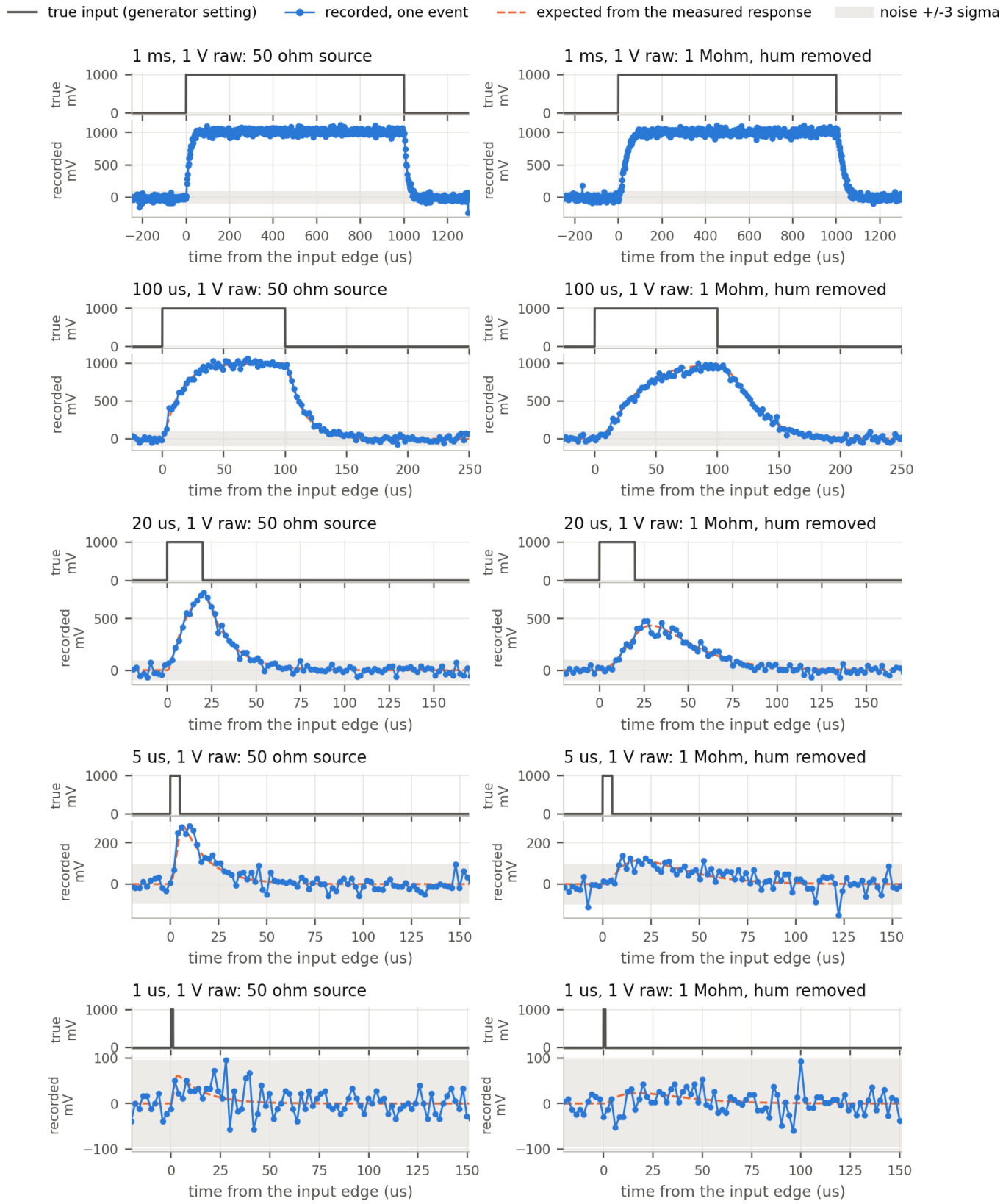
Generator at 0 V, so the board input sits behind 1 Mohm with nothing driving it. 60 Hz and its harmonics appear; at the 50 ohm source they were below 1 mV. How much hum a real source picks up depends on the wiring: leads, clips and their distance to mains wiring. It scales with the source impedance, so a 10 Mohm source would pick up several times more.





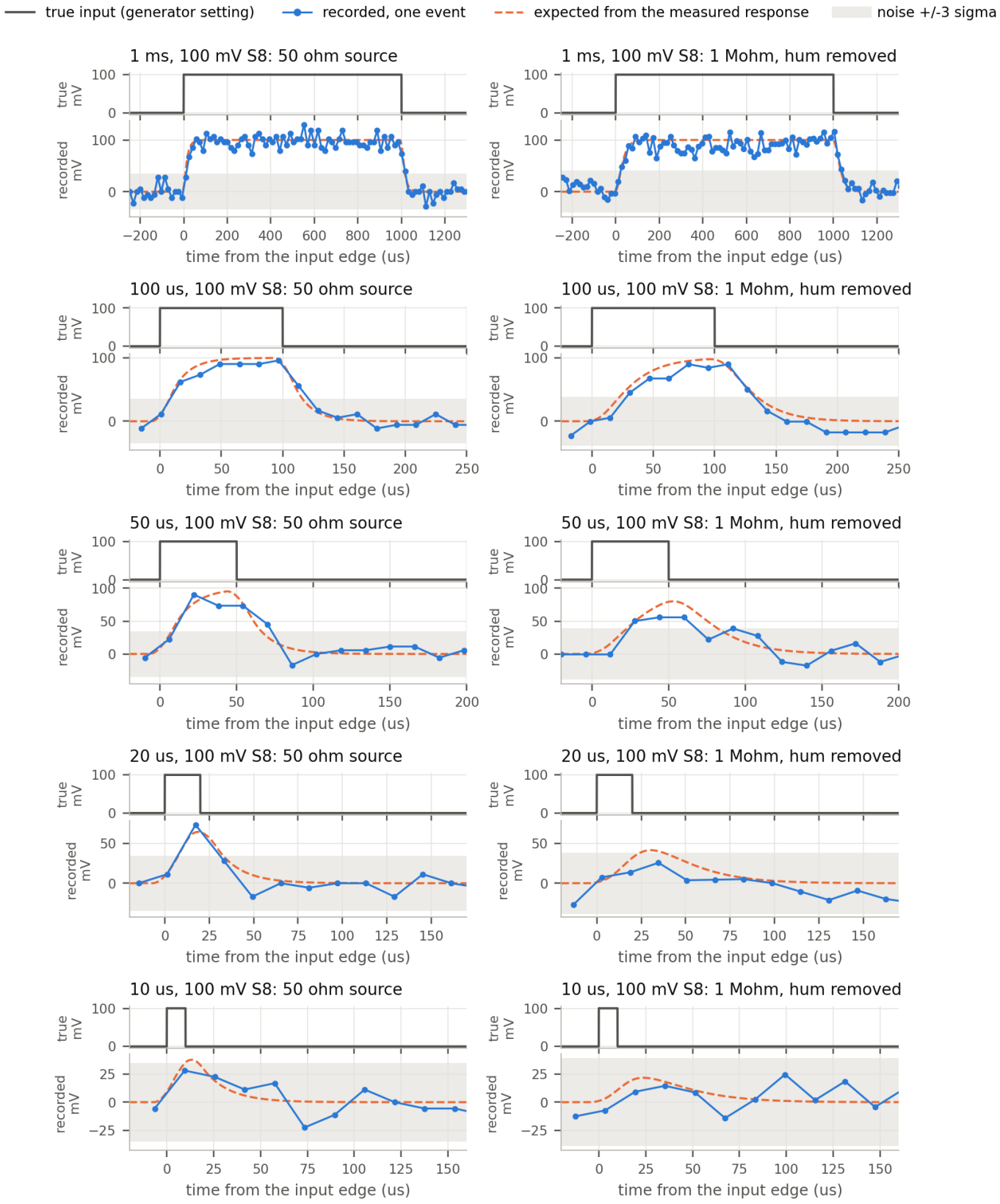
3. Pulse length at 1 V, 500 kS/s raw: 50 ohm vs 1 Mohm

Left the 50 ohm run, right 1 Mohm with the hum removed; same scales in each row. Long pulses still reach full height; their edges are slower. Short pulses come out smaller and wider.

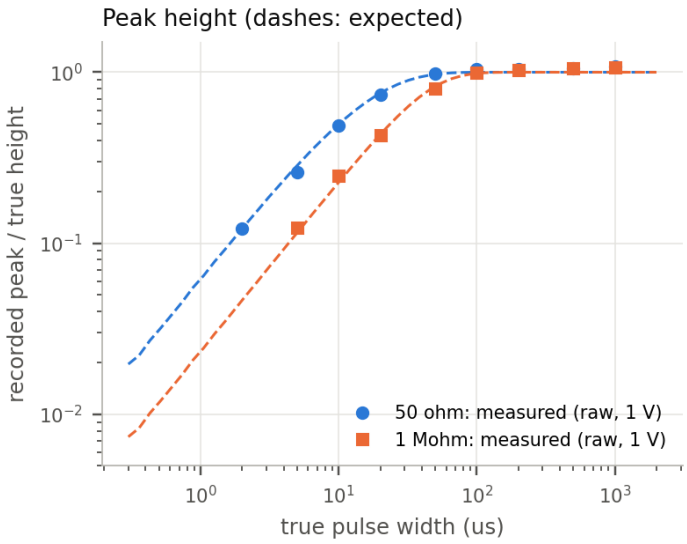


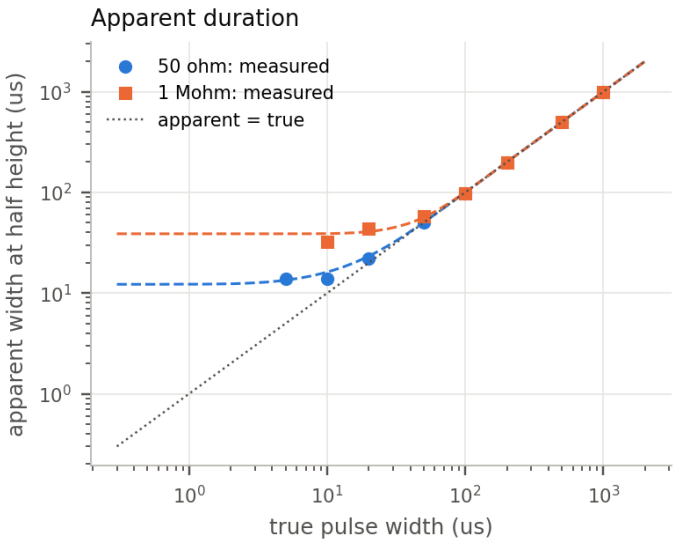
4. Weak pulses: 100 mV, SMOOTH 8

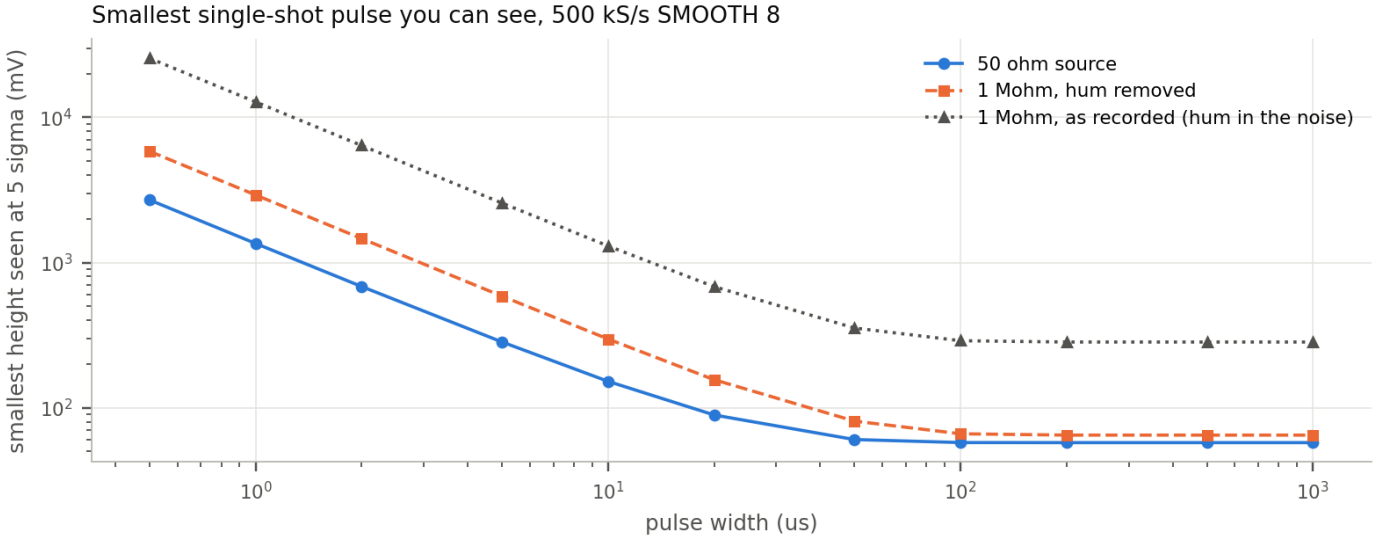
With the hum removed, the noise is close to the 50 ohm run (13.0 vs 11.7 mV), so long weak pulses are seen almost as well. The loss is at the short end, where the extra pole takes height away.



5. Height and apparent duration vs pulse length







6. Ringing (3-cycle bursts, 2 V peak, raw)

frequency	recorded / true, 50 ohm	recorded / true, 1 Mohm
2 kHz	0.98	0.96
10 kHz	0.77	0.59
30 kHz	0.46	0.22
100 kHz	0.19	0.04 (noise)
300 kHz	0.05 (noise)	0.01 (noise)

7. What it means for a real high-impedance source

- **Expect the extra pole to scale with the source resistance.** With ~18 pF at the input, 10 Mohm would put it near 867 Hz (time constant 184 us). Then even 1 ms events would be rounded, and anything under ~100 us would be recorded only as its area. The capacitance is the lever: a shorter, lower-capacitance lead to the source or a buffer at the source helps more than any setting.
- **Deal with the hum.** Twisted or shielded leads, with the shield on the input common, keep it out at the source. Otherwise, subtract a fitted 60/120/180 Hz in software as this report does, or remove a local baseline around each event. Either way, don't count on SMOOTH to remove it.
- **The offset shifts with source resistance.** -14 mV at 1 Mohm suggests ~14 nA of input current; at 10 Mohm that would be ~141 mV. Null it with the source connected, not with the input shorted.
- **Settings don't change.** 500 kS/s with SMOOTH 8 is still the best single-shot setting.

Not covered

- Only 1 Mohm was measured; the 10 Mohm numbers above are extrapolated from the capacitance and the offset shift.
- The hum figure is for this bench's wiring. A different lead layout will pick up more or less.

Reproducing this report

py frontend_characterize.py --board COM32 --label usb_1M_series
py transient_capture.py --board COM32 --label usb_1M
py series_r_report.py runs/<base>_transients_usb runs/<base>_usb_bench runs/<stamp>_transients_usb_1M runs/<stamp>_usb_1M_series 1e6

Data: 20261105-0754_transients_usb, 20261105-0617_usb_bench, 20261106-0739_transients_usb_1M, 20261106-0735_usb_1M_series.